World Ocean Science Congress 2026

23 – 26 February 2026
at CSIR-National Institute of Oceanography, Dona Paula, Goa, India

Automated Shoreline Extraction And Change Analysis Using Deep Learning And Dsas: A Case Study Of Mousuni Island, India

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ABSTRACT

Shoreline retreat poses a critical challenge to coastal resilience in monsoon-dominated marine environments, where shoreline evolution is controlled by complex interactions among wave climate, sediment transport, and coastal morphology. Mousuni Island in the western Indian Sundarbans is particularly vulnerable due to strong tidal currents, wave–tide interactions, and reduced sediment supply, resulting in enhanced hydrodynamic stress and accelerated shoreline instability. This study presents a comprehensive 35-year (1990–2025) assessment of shoreline change along Mousuni Island using an integrated framework combining geoinformatics, deep learning–based shoreline extraction, and the Digital Shoreline Analysis System (DSAS). Multi- temporal Landsat imagery was employed to implement and evaluate two semantic segmentation models, U-Net and DeepLabV3+, for automated shoreline delineation, with U-Net demonstrating superior boundary precision and spatial consistency across diverse coastal settings. Quantitative shoreline change analysis using DSAS-derived metrics, including Net Shoreline Movement (NSM), Shoreline Change Envelope (SCE), and Linear Regression Rate (LRR), reveals pronounced spatial variability in erosion and accretion patterns driven by monsoon-induced wave energy, sediment redistribution, and localized anthropogenic influences. The analysis revealed erosion as the dominant process, affecting nearly 90% of the island's perimeter. The mean shoreline retreat rate was −4.36 m yr^{-1}, with extreme erosion rates of up to −23.55 m yr^{-1} observed in the southern and northwestern high-risk zones. In contrast, limited accretion was identified in localized northeastern sectors, with a maximum shoreline advance of +2.81 m yr^{-1}.The study introduces a replicable framework that integrates historical erosion patterns with future risk projections to assess deltaic island vulnerability, providing actionable insights for nature-based solutions, strategic embankment design, and climate-resilient coastal management.

Keywords: Shoreline change, Coastal erosion, Monsoon dynamics, Deep learning, DSAS

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